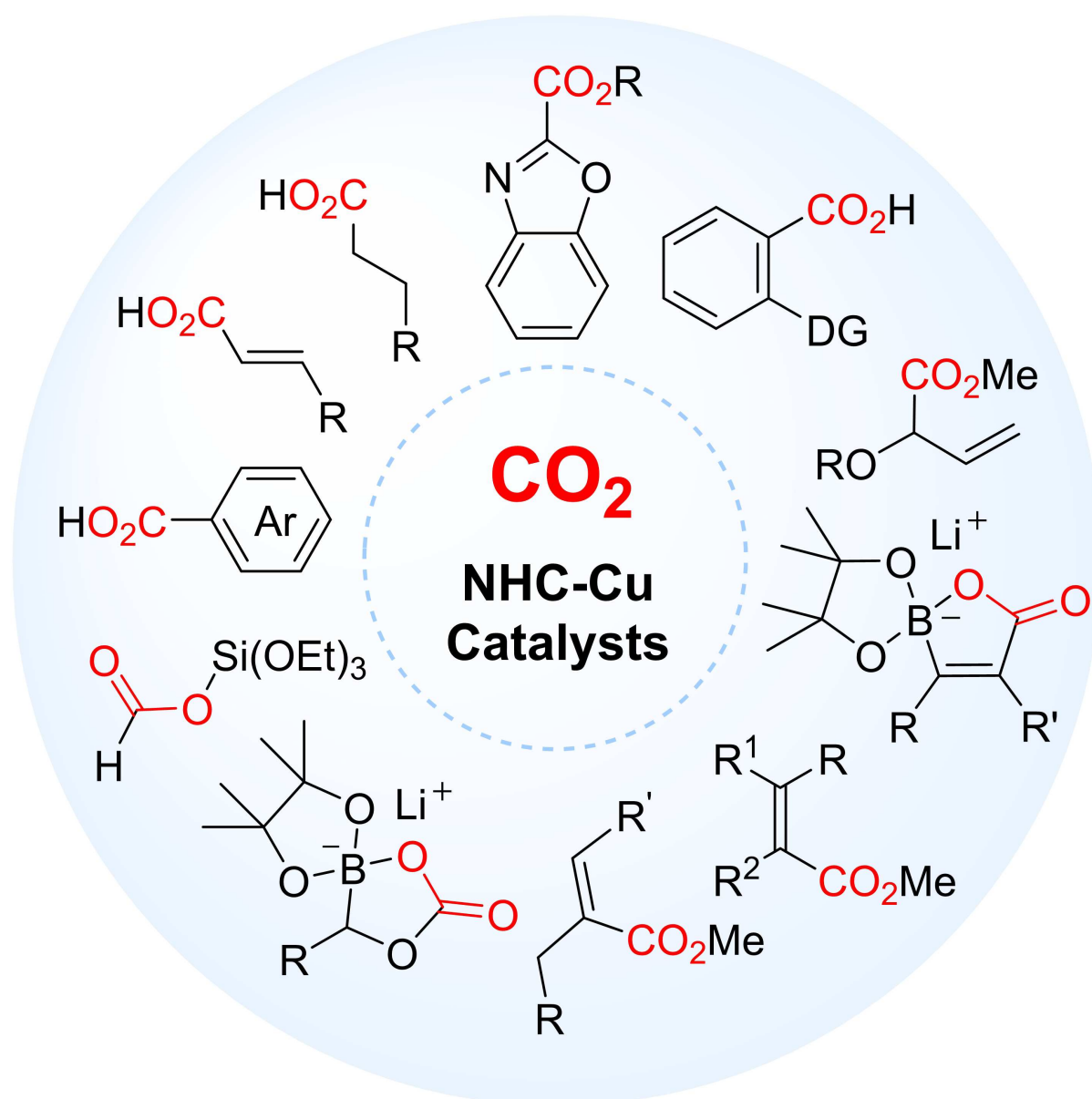


Carboxylation Reactions with Carbon Dioxide Using N-Heterocyclic Carbene-Copper Catalysts

Liang Zhang,^[a, b] Zhenghua Li,^[b] Masanori Takimoto,^[a, b] and Zhaomin Hou^{*[a, b]}



Abstract: The development of versatile catalyst systems and new transformations for the utilization of carbon dioxide (CO₂) is of great interest and significance. This Personal Account reviews our studies on the exploration of the reactions of CO₂ with various substrates by the use of N-heterocyclic carbene (NHC)-copper catalysts. The carboxylation of organoboron compounds gave access to a wide range of carboxylic acids with excellent functional group tolerance. The C–H bond carboxylation with CO₂ emerged as a straightforward protocol for the preparation of a series of aromatic carboxylic esters and butenoates from simple substrates. The hydrosilylation of CO₂ with hydrosilanes provided an efficient method for the synthesis of silyl formate on gram scale. The hydrogenative or alkylative carboxylation of alkynes, ynamides and allenamides yielded useful α,β -unsaturated carboxylic acids and α,β -dehydro amino acid esters. The boracarboxylation of alkynes or aldehydes afforded the novel lithium cyclic boralactone or boracarbonate products, respectively. The NHC-copper catalysts generally featured excellent functional group compatibility, broad substrate scope, high efficiency, and high regio- and stereoselectivity. The unique electronic and steric properties of the NHC-copper units also enabled the isolation and structural characterization of some key intermediates for better understanding of the catalytic reaction mechanisms.

Keywords: alkynes, carbon dioxide, carboxylation, copper catalysts, N-heterocyclic carbene

1. Introduction

The efficient utilization of sustainable feedstocks for the production of value-added chemicals is of great importance. In recent years, carbon dioxide (CO₂) has been attracting considerable attention for being used as an abundant, inexpensive, and renewable one-carbon (C1) building block in organic synthesis.^[1] Various catalyst systems, especially transition metal catalysts, have been developed for the transformation of CO₂ into carboxylic acids and their derivatives, which are privileged structural units in biologically active molecules and synthetically useful fine chemicals.^[2] Despite these progresses, the use of cheap and well-defined catalysts for achieving efficient and selective CO₂ conversion still represents a persistent challenge. On the other hand, the designing of new routes for incorporating CO₂ into valuable compounds with novel structures remains a formidable opportunity for the applicable transformation of CO₂.

We have developed a series of carboxylation reactions of CO₂ with various organic substrates by using N-heterocyclic carbene (NHC)-copper catalysts. The reaction generally proceeded through the facile insertion of CO₂ into an NHC-organocopper species generated *in situ* in a catalytic circle. Various types of carboxylic acids or esters have been synthesized from efficient and selective conversions of simple

and readily available starting materials. Remarkably, the boracarboxylation of alkynes and aldehydes with CO₂ and diboron reagents could be selectively achieved to afford unique lithium cyclic boralactone and boracarbonate products with novel molecular structures, respectively. The NHC-copper platform enabled isolation and structural characterization of some key intermediates, thus providing important insights into the mechanistic details. This account mainly summarizes our studies on NHC-copper-catalyzed carboxylation reactions with CO₂. Related studies by other groups are also described briefly.

2. Catalytic Carboxylation of Organoboron Compounds

2.1. Carboxylation of Aryl- and Alkenylboronic Esters

Due to its high thermodynamic stability and low reactivity, the use of CO₂ as a C1 building block for the formation of C–C bonds usually requires highly nucleophilic organometallic reagents, such as alkyllithium compounds and Grignard reagents. However, sensitive functional groups are generally not tolerated in the presence of these strong organometallic nucleophiles.

Transition-metal-catalyzed addition of carbon nucleophiles, in particular, less nucleophilic organoboron compounds, to CO₂ has attracted much attention.^[3] Iwasawa and co-workers have reported the Rh(I)-catalyzed carboxylation of aryl- and alkenylboronic esters with CO₂, which is potentially useful for the synthesis of valuable carboxylic acid derivatives because of the accessibility of various functionalized organoboronic esters.^[4] However, the Rh catalyst systems showed only limited tolerance toward some reactive functional groups, such as bromo, iodo, and vinyl groups. Besides, information

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about the active catalyst species and the reaction mechanism was not available because of the complexity of the catalyst systems.

In 2008, we have reported an excellent NHC–Cu(I) catalyst system for the carboxylation of aryl- and alkenylboronic esters with CO₂, which not only showed higher functional group compatibility, but also afforded structurally characterizable active catalyst species.^[5]

Systematic screening revealed that the isolated NHC–Cu(I) chloride complex [(IPr)CuCl]^[6] (Figure 1) shows high activity for the carboxylation reaction. The use of a less steric demanding NHC ligand, such as IMes·HCl, led to a lower yield of the carboxylation product. The reaction could not proceed in the absence of CuCl or an NHC ligand or KO^tBu under the same conditions.

Various aryl- and alkenylboronic esters could be used as suitable substrates under the optimized reaction conditions (Table 1). The efficiency of these reactions was not affected by either electron-donating or electron-withdrawing groups on the substrates. Notably, a variety of reactive functional groups,

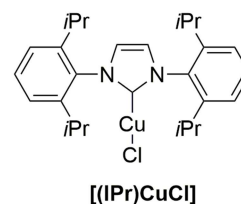


Figure 1. Structure of NHC-copper catalyst [(IPr)CuCl].

such as vinyl, epoxy, carbonyl, halide, and propargyl ether, were tolerated. The reactions of boronic esters with sterically demanding aryl groups, heteroaromatic derivatives, and even alkenyl substituents took place selectively to give the corresponding carboxylation products in high yields.

Sequential stoichiometric reactions of NHC-copper complexes were examined to gain some information on the reaction mechanism (Scheme 1). Under a N₂ atmosphere, [(IPr)CuCl] reacted with 1 equivalent of KO^tBu in THF at room temperature, giving a known copper alkoxide complex



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Zhenghua Li obtained his PhD from Katholieke Universiteit Leuven, Belgium in 2016 under the supervision of Professor Erik Van der Eycken. At the same year, he started his postdoctoral research in Professor Zhaomin Hou's group at RIKEN. His research interests focus on the development of metal-catalyzed multicomponent reactions for valuable heterocycles and fine chemicals.



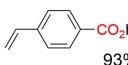
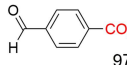
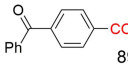
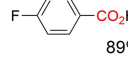
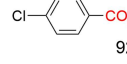
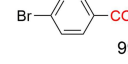
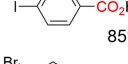
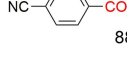
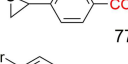
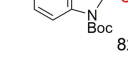
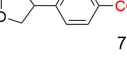
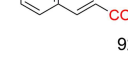
Masanori Takimoto received his PhD from Hokkaido University in 1998. He conducted postdoctoral research at University of Toronto (1989–1999, with Professor Mark Lautens). In April 1999, he was appointed as an assistant professor of Hokkaido University and joined Professor Miwako Mori's group. He moved to



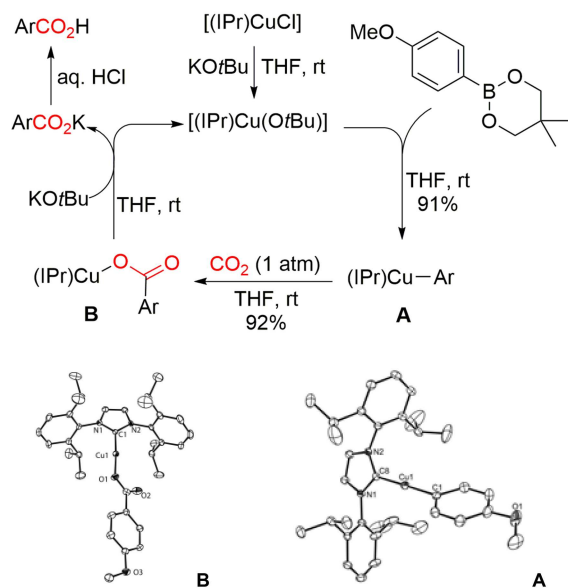
Professor Zhaomin Hou's group in RIKEN as a tenured Research Scientist in 2005 and was appointed to a Senior Research Scientist in 2008. His current research interests include organometallic chemistry, synthetic organic chemistry, and their applications.

Zhaomin Hou received his PhD from Kyushu University in 1989. He conducted postdoctoral research at RIKEN (1989–1991, with Professor Hiroshi Yamazaki) and University of Windsor (1991–1993, with Professor Douglas Stephan). He joined RIKEN as a tenured Research Scientist in 1993 and was appointed to Chief Scientist and Director of Organometallic Chemistry Laboratory at RIKEN in 2002. He now also serves as a Deputy Director of RIKEN Center for Sustainable Resource Science which was inaugurated in April 2013. His current research interests include the activation and functionalization of small molecules and inert chemical bonds, and development of new catalysts for more efficient, selective chemical synthesis, including asymmetric C–H transformation and copolymerization of non-polar and polar olefins by organo rare-earth catalysts.

Table 1. [(IPr)CuCl]-catalyzed carboxylation of aryl- and alkenylboronic esters with CO₂.^[a]

$\text{R-B} \begin{array}{c} \diagup \text{O} \diagdown \\ \diagdown \text{O} \diagup \end{array} + \text{CO}_2 \xrightarrow[\text{THF, reflux, 24 h}]{\begin{array}{c} \text{[(IPr)CuCl]} (1 \text{ mol}\%) \\ \text{KOtBu} (1.05 \text{ mmol}) \\ 1 \text{ atm} \end{array}} \xrightarrow{\text{H}^+} \text{R-CO}_2\text{H}$		
		
93%	97%	89%
		
89%	92%	99%
		
85%	88%	77%
		
82%	73%	92%

[a] Reaction conditions: [(IPr)CuCl] (1 mol%), boronic ester (1 mmol), KOtBu (1.05 mmol), CO₂ (1 atm), THF (5 mL), reflux, 24 h. Isolated yields.

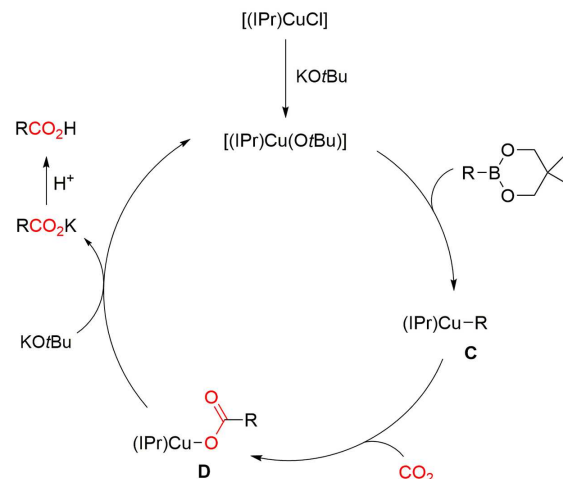


Scheme 1. Stoichiometric reactions of copper complexes.

[(IPr)Cu(OtBu)].^[7] Further reaction of [(IPr)Cu(OtBu)] with an arylboronic ester generated the corresponding arylcopper complex **A** in 91% isolated yield. Introduction of CO₂ gas into a THF solution of **A** at room temperature afforded the carboxylate complex **B** in 92% isolated yield. Treatment of **B** with 1 equivalent of KOtBu in THF regenerated [(IPr)Cu(OtBu)], releasing potassium 4-methoxybenzoate as a white precipitate. Hydrolysis of the obtained potassium salt gave 4-

methoxybenzoic acid quantitatively. It was found that [(IPr)Cu(OtBu)], the aryl complex **A**, and the carboxylate **B** also showed high catalytic activity for the current carboxylation reactions.

A possible catalytic cycle for the NHC-copper catalyzed carboxylation of organoboronic esters with CO₂ was proposed (Scheme 2). The organocopper complex **C** would be generated



Scheme 2. A possible mechanism for the [(IPr)CuCl]-catalyzed carboxylation of aryl- and alkenylboronic esters with CO₂.

from the reaction of an organoboronic ester with the alkoxide complex [(IPr)Cu(OtBu)], which was formed from the metathesis reaction between [(IPr)CuCl] and KOtBu. The following insertion of CO₂ into **C** should afford the carboxylate intermediate **D**, which reacted with KOtBu to release RCO₂K and regenerate [(IPr)Cu(OtBu)]. After hydrolysis with HCl, the potassium salt was converted to the final product RCO₂H.

A CuI/bisoxazoline catalyst system reported by Iwasawa also showed good functional group tolerability for the carboxylation of organoboron compounds with CO₂.^[8]

2.2. Carboxylation of Alkylboranes

Compared to the carboxylation of aryl- and alkenylboronic esters, the reaction of alkylboranes with CO₂ is challenging because the generation of a transition metal alkyl intermediate showing appropriate stability and high activity towards CO₂ from an alkylborane is more difficult. Besides, although alkylboranes have been widely used as substrates in transition metal-catalyzed cross-coupling reactions,^[9] the isolation and characterization of the intermediate in the transmetalation of an organoboron compound to a transition metal is rare.^[10] Thus, the information on the interaction between an

Table 2. [(IPr)CuCl]-catalyzed carboxylation of terminal alkenes with CO₂.^[a]

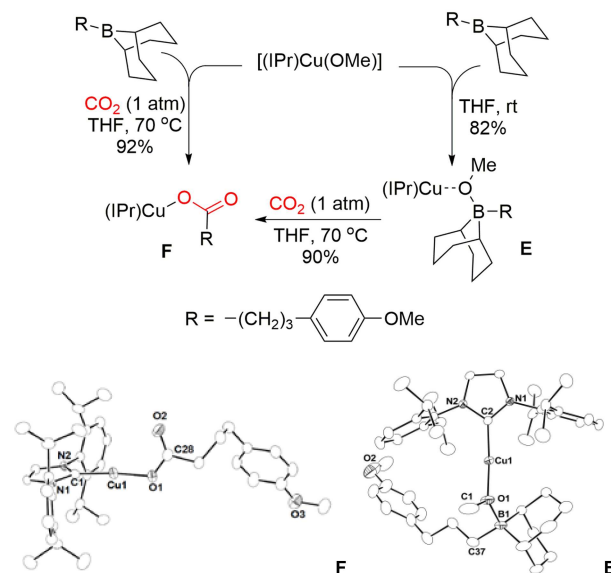
 97%	 92%	 94%
 91%	 95%	 99%
 90%	 91%	 82%
 97%	 81%	 92%
 70%	 65%	

[a] Isolated yields.

alkylborane and a transition metal species remained very limited.

We have reported that [(IPr)CuCl] can serve as an excellent catalyst for the carboxylation of alkylboranes, which were easily prepared *in situ* by hydroboration of terminal alkenes with 9-borabicyclo[3.3.1]nonane (9-BBN).^[11] The reactions of various functionalized alkylboranes with CO₂ were investigated by using [(IPr)CuCl] as a catalyst and MeOLi as a base (Table 2). A variety of reactive functional groups, such as propargyl, carbonyls, vinyl bromide, and halides (I, Br, Cl, F) are all compatible with the present reaction conditions. Moreover, synthetically useful protecting groups, such as silyl ether, *N*-tert-butoxycarbonyl, benzyl ether, and isopropylidene can also be tolerated. Alkylboranes containing heteroaromatic motifs are also found to be suitable substrates. The reactions of a terminal alkene bearing a tertiary substituent and even styrene with 9-BBN and CO₂ afforded the corresponding products in 70 % and 65 % yields, respectively.

In order to elucidate the reaction mechanism of the catalytic cycle, several sequential stoichiometric reactions of NHC-copper complexes were examined (Scheme 3). When [IPrCu(OMe)]^[12] was mixed with an alkylborane in THF at room temperature, an adduct complex **E** was generated instantly. X-ray crystallographic studies revealed that **E** is formed by the interaction between the oxygen atom in [IPrCu(OMe)] and the boron atom in alkylborane. When a solution of **E** in THF was exposed to a CO₂ atmosphere at 70 °C, the carboxylate complex **F** was isolated in 90 % yield. Alternatively, **F** could also be obtained by the one-pot reaction of [IPrCu(OMe)] with alkylborane under a CO₂ atmosphere. It


Scheme 3. Stoichiometric reactions of copper complexes.

was found that [IPrCu(OMe)], **E** and **F** all showed high catalytic activity for the carboxylation reaction under the standard conditions. The isolation of the novel copper-methoxide/alkylborane adduct complex **E** has provided important information of the mechanistic details. Although the interaction between a basic metal alkoxide or hydroxide with an organoboron substrate has been thought critically important in related transition metal-catalyzed cross-coupling reactions,^[13] the structure of **E** demonstrates unambiguously that such bonding interaction does exist for the first time.

An analogous carboxylation of alkylboranes with CO₂ catalyzed by a CuOAc/1,10-phenanthroline catalyst system was reported by Sawamura and co-workers.^[14]

3. Carboxylation of C–H Bonds

3.1. Direct Carboxylation of Heteroaromatic C–H Bonds

The C–C bond formation reaction with CO₂ usually requires the use of pre-activated substrates, such as organometallic reagents or organic halides. The direct carboxylation of C–H bonds with CO₂, which represents the most attractive and atom-economic method for the synthesis of carboxylic acids, remained a challenge. Heterocyclic carboxylic acids and their derivatives are privileged structural motifs in medicinally important molecules and synthetically useful chemicals.^[15] We have found that [(IPr)CuCl] can be used as an efficient catalyst for the direct carboxylation of aromatic heterocyclic C–H

Table 3. Ligand effect of Cu-catalyzed carboxylation of benzoxazole with carbon dioxide.^[a]

Entry	L	Yields (%)
1	IPr-HCl	69
2	SIPr-HCl	70
3	IMes-HCl	62
4	SIMes-HCl	70
5	PPh ₃	7
6	1,10-Phen	21
7	L-Proline	5
8	TMEDA	trace

[a] Reaction conditions: CuCl (10 mol%), L (10 mol%), benzoxazole (1.0 mmol), KOtBu (1.1 mmol), CO₂ (1 atm), THF (5 mL), 80 °C, 14 h, then THF was evaporated; DMF (5 mL), C₆H₁₃I (2.0 mmol), 80 °C, 5 h. Isolated yields.

Table 4. [(IPr)CuCl]-catalyzed carboxylation of aromatic heterocycles with CO₂.^[a]

[a] Isolated yields.

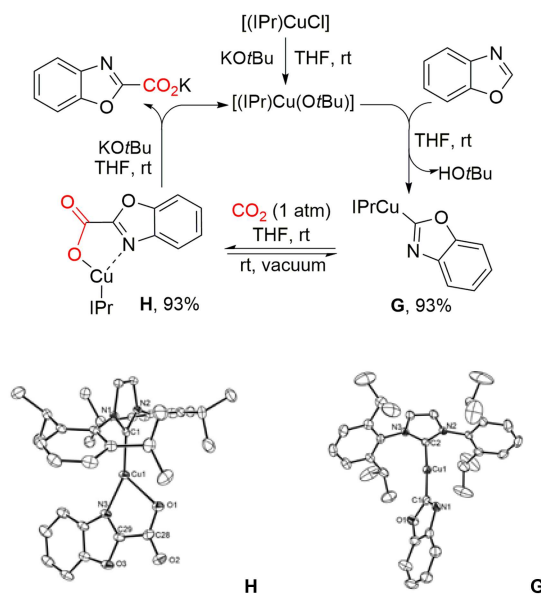
bonds with CO₂, which afford the corresponding heterocyclic carboxylic esters after treatment with an alkyl iodide.^[16]

We systematically screened different kinds of ligands for the copper-catalyzed carboxylation of benzoxazole with CO₂ (Table 3). It was found that the combinations of CuCl with N-heterocyclic carbene ligands such as IPr-HCl, SIPr-HCl, IMes-HCl or SIMes-HCl were all effective catalyst systems. Other phosphine- or nitrogen-based ligands, such as PPh₃,

1,10-phenanthroline, L-proline, and tetramethylethylenediamine (TMEDA) were much less effective. We then examined the reaction of benzoxazole with CO₂ by the use of [(IPr)CuCl] as a catalyst, which was found to be the best of choice. The reaction was carried out in the presence of 5 mol% of [(IPr)CuCl] and 1.1 equivalent of KOtBu under 1 atm of CO₂ in THF at 80 °C for 14 h. In an one-pot fashion, the resulting potassium benzoxazole-2-carboxylate salt was treated with MeI, EtI or *n*-C₆H₁₃I in DMF at 80 °C for 5 h, affording the corresponding benzoxazole-2-carboxylate esters in high isolated yields.

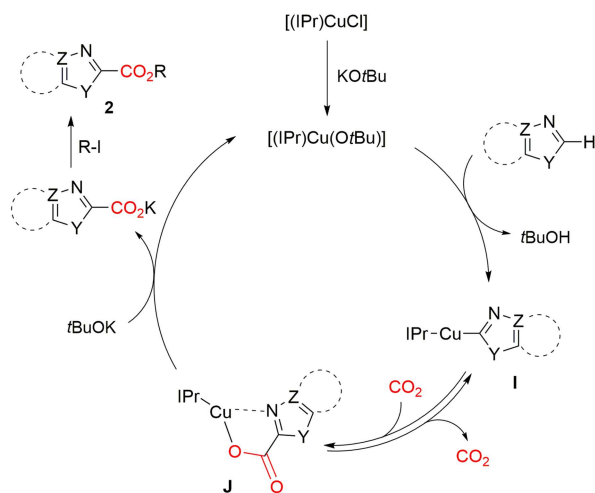
The carboxylation reactions of various aromatic heterocyclic compounds were examined in the presence of [(IPr)CuCl] as a catalyst and *n*-C₆H₁₃I as an alkylating agent (Table 4). Both 5-methylbenzoxazole and 6-methylbenzoxazole reacted with CO₂ smoothly to give the corresponding esters in 85% and 87% isolated yields, respectively. However, the reaction of 4-methylbenzoxazole gave a lower yield (50%) of the corresponding ester, which may due to the steric influence of the methyl group to the adjacent heterocyclic ring. Chloride, bromide and nitro groups on benzoxazole substrates are compatible with the reaction conditions. 5-Arylated benzoxazole substrates bearing either electron-donating (e.g., -OMe) or electron-withdrawing (e.g., -CF₃, -CN) functional groups could give the corresponding functionalized esters in high yields. In the case of 5-(methylbenzoate)-substituted benzoxazole, the expected methyl ester product was obtained in 52% isolated yield, along with a *tert*-butyl ester byproduct which was formed by transesterification with *in situ* generated *t*BuOH. Heterocycles with weaker acidity such as *N*-methylbenzimidazole and 1,3,4-oxadiazole afforded the expected products in lower yields.^[17] Similarly, only trace amounts of the carboxylation products were observed in the reactions of benzothiazole, 4-phenyloxazole, and benzofuran.

In a series of stoichiometric reactions (Scheme 4), [(IPr)Cu(OtBu)] firstly reacted with benzoxazole to give the benzoxazolylcopper complex **G** in 93% isolated yield. On exposure of the THF solution of **G** to 1 atm of CO₂ at room temperature, the carboxylate complex **H** was isolated in 93% yield. Interestingly, an X-ray crystallographic diffraction study of **H** revealed that the benzoxazolylcarboxylate unit is bonded to the Cu atom to form a five-membered ring through a carboxylate oxygen atom and the nitrogen atom. Decarboxylation took place when **H** was kept under vacuum overnight or under a N₂ atmosphere for 1 week at room temperature, affording **G** almost quantitatively. Treatment of the carboxylate complex **H** with 1 equivalent of KOtBu in THF regenerated [(IPr)Cu(OtBu)] and afforded the potassium carboxylate salt quantitatively. Complexes [(IPr)Cu(OtBu)], **G** and **H** all showed high catalytic activity for the carboxylation of benzoxazole under the standard conditions.



Scheme 4. Stoichiometric reactions of copper complexes.

In a proposed reaction mechanism for the current copper-catalyzed carboxylation of aromatic heterocycles with CO_2 (Scheme 5), the copper alkoxide complex $[(\text{IPr})\text{Cu}(\text{O}t\text{Bu})]$ firstly reacts with a heterocycle substrate to give the organo-copper intermediate **I** through deprotonation of the aromatic heterocyclic C–H bond. The following insertion of CO_2 into the Cu–C bond in **I** yields the copper carboxylate **J**, which could regenerate **I** through a facile decarboxylation process.



Scheme 5. Proposed mechanism for the $[(\text{IPr})\text{CuCl}]$ -catalyzed carboxylation of aromatic heterocycles with CO_2 .

Further reaction of **J** with $\text{KO}t\text{Bu}$ regenerates $[(\text{IPr})\text{Cu}(\text{O}t\text{Bu})]$ and releases the potassium carboxylate, which affords the ester product after treatment with alkyl iodide.

The similar C–H carboxylation of benzoxazole and benzothiazole derivatives could also be achieved in the presence of 1,2,3-triazol-5-ylidene copper(I) complexes.^[18] The carboxylation of acidic aromatic C–H bond bonds with CO_2 by the use of a NHC–Cu complex $[(\text{IPr})\text{CuOH}]$ ^[19] or a NHC–Au complex $[(\text{IPr})\text{AuOH}]$ ^[20] have also been reported.

3.2. Formal C–H Carboxylation of Aromatic Compounds

The mixed alkylamido lithium aluminate compound $i\text{Bu}_3\text{Al}(\text{TMP})\text{Li}$ ($\text{TMP} = 2,2,6,6\text{-tetramethylpiperidide}$) has been reported to selectively deprotonate certain functionalized aromatic compounds, affording the corresponding arylaluminum species after directed *ortho*-metalation.^[21] We have found that NHC-copper catalysts enable the carboxylation of organo-boron compounds and heteroaromatics with CO_2 .^[5,11,16] Inspired by these results, we successfully combined the deprotonative *ortho*-aluminum of aromatics by $i\text{Bu}_3\text{Al}(\text{TMP})\text{Li}$ with the NHC-copper catalyzed carboxylation of the resulting arylaluminum species with CO_2 , which represents a useful method for the selective C–H carboxylation of various aromatic compounds.^[22,23]

It was found that the arylaluminum species formed by the deprotonative *ortho*-aluminum of *N,N*-diisopropylbenzamide with $i\text{Bu}_3\text{Al}(\text{TMP})\text{Li}$ did not react with CO_2 (1 atm) alone at room temperature. The carboxylation of the *in situ* generated arylaluminum species was then investigated by the use of NHC-copper complexes as the catalysts.^[22] In the presence of 5 mol% of $[(\text{IPr})\text{CuCl}]$ and 5 mol% $\text{KO}t\text{Bu}$, the corresponding carboxylic acid product was obtained in 92 % yield. Under the optimized reaction conditions, the carboxylation of various substituted benzene derivatives was examined (Table 5). In the case of *t*Bu- and halide-substituted *N,N*-diisopropylbenzamides, the reactions occur selectively to give the *ortho* C–H bond carboxylation products in high yields. The similar carboxylation of benzonitrile also took place selectively, affording 2-cyanobenzoic acid in 84 % yield. Anisole derivatives such as 4-methylthioanisole and 4-allylanisole are also applicable to this method, yielding the products formed by carboxylation at the *ortho*-position of the methoxy group. By the use of same protocol, heteroaromatic substrates like benzofuran, benzothiophene, and *N*-Boc-indoles, could be selectively converted to the corresponding C2 carboxylation products in high yields.

3.3. C–H Carboxylation of Allyl aryl Ethers

The efficient and selective preparation of 2-aryloxy butenoic acid derivatives is of much interest and importance.^[24]

Table 5. Deprotonative carboxylation of aromatic compounds.^[a]

$(\text{het})\text{Ar}-\text{H} \xrightarrow[\text{THF, -78 } ^\circ\text{C to rt, 3 h}]{2.0 \text{ equiv } i\text{Bu}_3\text{Al}(\text{TMP})\text{Li}}$		$\xrightarrow[\text{THF, rt, 20 h}]{5 \text{ mol\% } [(i\text{Pr})\text{CuCl}], 5 \text{ mol\% KOtBu, 1 atm CO}_2}$		$(\text{het})\text{Ar}-\text{CO}_2\text{H/Me}$			
	92%		88%		74%		95%
	84%		90%		50%		72%
	90%		94%		86%		73%

[a] Isolated yields.

Table 6. Cu-catalyzed allylic C–H carboxylation with CO₂.^[a]

$\text{RO}-\text{CH}_2-\text{CH}=\text{CH}_2 \xrightarrow[\text{THF, -78 } ^\circ\text{C to rt, 3 h}]{2.0 \text{ equiv } i\text{Bu}_3\text{Al}(\text{TMP})\text{Li}}$		$\xrightarrow[\text{THF, 0 } ^\circ\text{C, 2 h}]{10 \text{ mol\% } [(i\text{Mes})\text{Cu}(\text{O}t\text{Bu})], 1 \text{ atm CO}_2}$		without DBU	10 mol% DBU
	88%		90%		75%
	87%		89%		80%
	70%		65%		65%
	70%		65%		70%
	87%		86%		82%
	82%		76%		78%
	82%		78%		78%

[a] Isolated yields. TBDMS = *tert*-butyldimethylsilyl.

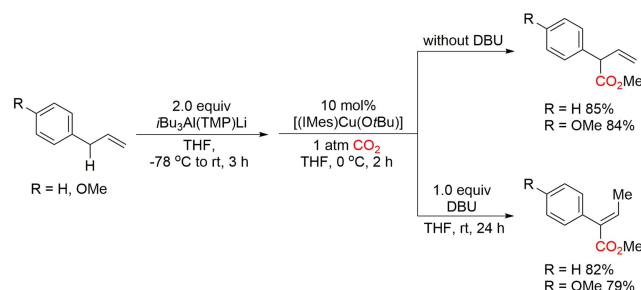
Previously, Uchiyama and co-workers reported that the aluminum ate bases such as *i*Bu₃Al(TMP)Li can selectively deprotonate the allylic C–H bond of allyl aryl ethers.^[25] Inspired by this report and our works on copper-catalyzed carboxylation with CO₂, we developed the NHC-copper catalyzed carboxylation of aryloxy allyl aluminum species

generated by the deprotonative C–H aluminatation of allyl aryl ethers with *i*Bu₃Al(TMP)Li.^[26,27] This reaction provides an efficient and selective method for the synthesis of 2-aryloxy butenoates using CO₂ as a C1 building block.

The allyl aluminum species were generated *in situ* from the deprotonative aluminatation of allyl phenyl ether with *i*Bu₃Al(TMP)Li. The subsequent carboxylation with CO₂ was examined by the use of various NHC-copper complexes as catalysts.^[26] In the presence of 10 mol% of [(IMes)Cu(O*t*Bu)], the carboxylation product methyl 2-phenyloxy-3-butenate was obtained in 91 % yield. It was found that various aryl allyl ethers are suitable substrates for this carboxylation reaction (Table 6, above the dashed line). Functional groups on the phenyl ring of aryl allyl ether, such as *m*-NMe₂, *p*-TBDMSO, *p*-CF₃, and halides are compatible with the reaction conditions, affording the corresponding products in high yields selectively. The reactions of aryl allyl ethers bearing a disubstituted allyl group could also give the expected products in slightly lower yields. Besides aryl allyl ethers, this protocol could also be applied to the carboxylation of propargylic C–H bond of phenyl propargyl ether and allylic C–H bond of carbamic acid allyl esters.

It was also found that in the presence of 10 mol% of DBU, isomerization of the double bond proceeded smoothly at room temperature to give 2-aryloxy-(*Z*)-2-butenate in a regio- and stereoselective manner. This protocol was applicable to various aryl allyl ether substrates (Table 6, below the dashed line).

These protocols were also applicable to less acidic allylbenzenes. The carboxylation of the benzylaluminum species that were prepared *in situ* from the corresponding allylbenzenes took place efficiently under standard conditions, affording the corresponding methyl 2-aryl-3-butenates in high yields. When the carboxylation reaction mixtures were treated with 1.0 equivalent of DBU, methyl (*E*)-2-aryl-2-butenates were obtained in high yields *via* the isomerization of methyl 2-aryl-3-butenates (Scheme 6).



Scheme 6. Carboxylation of allylbenzenes.

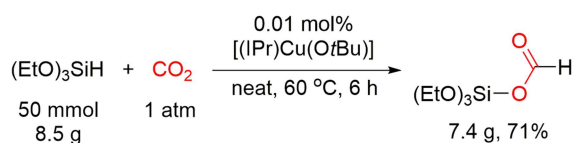
4. Hydrosilylation of CO₂

Various transition metal catalysts have been reported for the hydrogenation of CO₂ with H₂ for the synthesis of formic acid.^[28] Because this transformation is thermodynamically unfavorable, it generally requires relatively high pressure of CO₂ and H₂. Alternatively, the hydrosilylation of CO₂ with a hydrosilane is thermodynamically favorable, which could serve as a promising route for the preparation of silyl formates from CO₂.^[29] However, the catalytic activities of reported transition metal catalysts for hydrosilylation of CO₂ need to be improved.

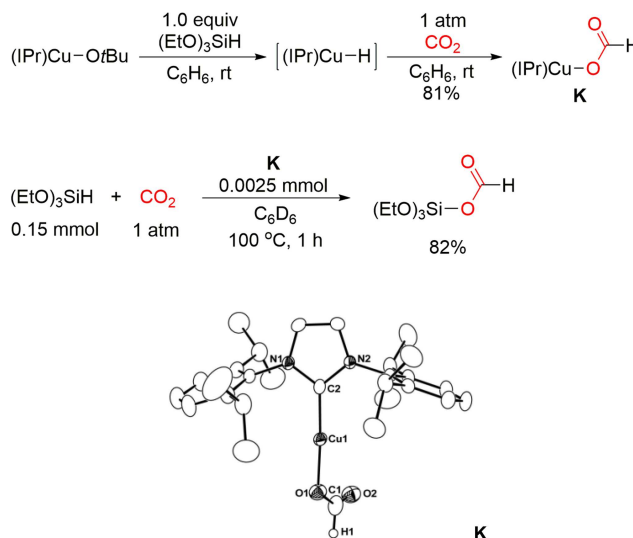
It was reported that the combination of $\text{Cu}(\text{OAc})_2 \cdot \text{H}_2\text{O}$ / 1,2-bis(diphenylphosphino)benzene efficiently catalyze the one-pot synthesis of formic acid from CO_2 and hydrosilanes.^[30] Because silyl formates can be used as formylating agents in organic synthesis and as a building block for silicon-based materials, the exploration of efficient catalyst systems for the direct synthesis of silyl formates from the hydrosilylation of CO_2 is of great significance. We have reported a copper-catalyzed hydrosilylation of CO_2 for the synthesis of silyl formate in multi-gram quantity under mild conditions.^[31] It was found that the hydrosilylation of CO_2 (1 atm) with $(\text{EtO})_3\text{SiH}$ (0.15 mmol) at 100°C was completed within 1 h in the presence of 0.0025 mmol of $[(\text{IPr})\text{Cu}(\text{O}i\text{Bu})]$. The less sterically demanding catalyst $[(\text{IMes})\text{Cu}(\text{O}i\text{Bu})]$ gave a lower yield under the same conditions. When the hydrosilylation reaction was carried out at 60°C , high conversion of $(\text{EtO})_3\text{SiH}$ was also achieved in 1 h. Even at room temperature, the reaction could be completed in a longer reaction time (10 h).

To show the practical usefulness of the current reaction, the hydrosilylation of CO₂ with (EtO)₃SiH on a 50 mmol scale was carried out under neat conditions with 0.01 mol% of [(IPr)Cu(OtBu)] (Scheme 7). Triethoxysilyl formate was generated in 75 % yield in 6 h, showing a TON of 7489 and a TOF of 1248 h⁻¹. The resulting triethoxysilyl formate product was readily isolated in 71 % yield by distillation.

As reported previously, the reaction of [(IPr)Cu(O*t*Bu)] with (EtO)₃SiH at room temperature generated instantly a copper hydride complex.^[7] Its *in situ* reaction with CO₂ at room temperature afforded the corresponding copper formate complex [(IPr)CuO₂CH] **K** in 81% yield (Scheme 8). The



Scheme 7. [(IPr)Cu(O*t*Bu)]-catalyzed hydrosilylation of CO₂ with hydrosilanes.



Scheme 8. Reactions of copper complexes with a hydrosilane and CO₂.

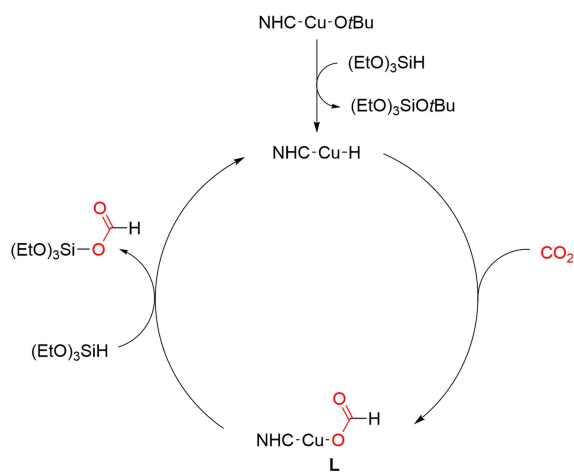
instant formation of [(IPr)CuH] was confirmed by ^1H NMR analysis when adding 1.0 equivalent of $(\text{EtO})_3\text{SiH}$ into a solution of complex **K** in benzene. The isolated formate complex **K** was also proved to be an efficient catalyst for the hydrosilylation of CO_2 . As revealed by X-ray diffraction study, the formate unit in complex **K** is connected to the Cu atom in a η^1 -bonding mode. In contrast, the formate unit is bonded to the Cu atom through both two oxygen atoms in the triphosphine-ligated copper formate complexes.^[32] The η^1 -bonding mode in complex **K** may due to the unique steric hindrance and electron donating property of the NHC ligand, which may account for the high activity of this catalytic system.

In a possible reaction mechanism (Scheme 9), the initial reaction of NHC-copper alkoxide complex with $(\text{EtO})_3\text{SiH}$ generates an active copper hydride. The following insertion of CO_2 into the Cu–H bond affords the corresponding copper formate complex **L**. The final reaction of $(\text{EtO})_3\text{SiH}$ with the copper formate **L** releases the desired silyl formate product and regenerates the copper hydride catalyst.

5. Carboxylation of Alkynes and Allenes

5.1. Boracarboxylation of Alkynes

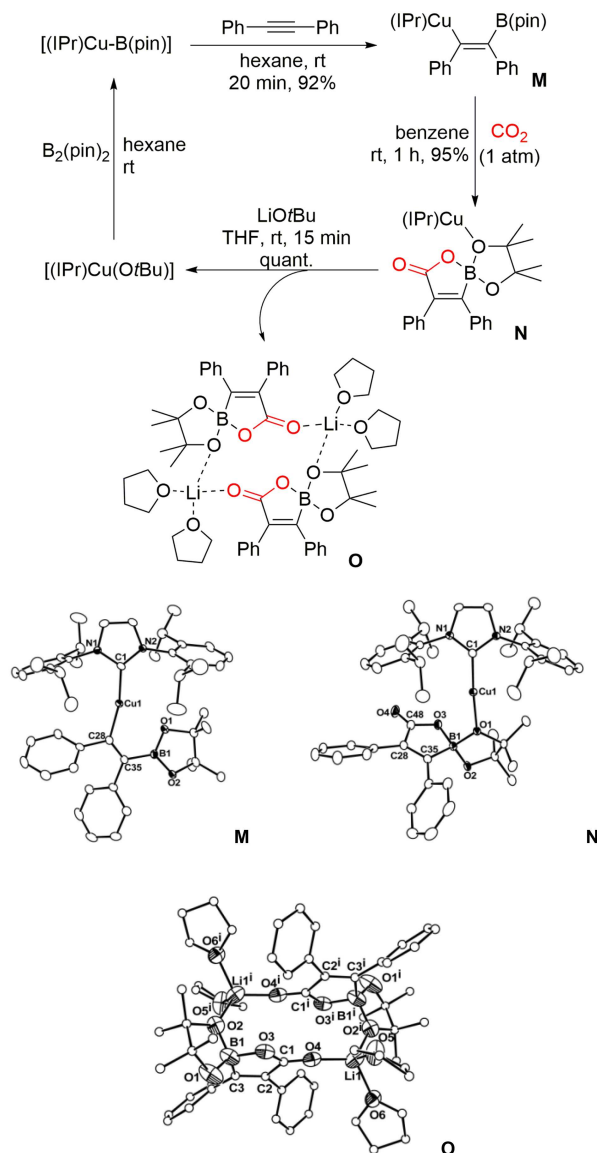
Recently, transition metal-catalyzed carboxylations of alkynes with CO₂, such as hydrocarboxylation, alkylative/arylate carboxylation, and double carboxylation, have been developed as a straightforward protocol for the synthesis of α,β -unsaturated carboxylic acids.^[2e,k,33] On the other side, the heterocarboxylation of alkynes with CO, can result in the



Scheme 9. Proposed reaction mechanism of catalytic hydrosilylation.

simultaneous addition of both a heteroatom unit and a carboxylate group across the C–C triple bond, which emerges as a powerful method for the preparation of multifunctionalized alkenes. Previously, only a few examples of the stoichiometric heterocarboxylation of highly reactive substrates were reported.^[34] The catalytic heterocarboxylation of simple unsaturated substrates with CO₂ is of great significance. We have reported the first catalytic boracarboxylation of alkynes with a diboron reagent and CO₂ by the use of a NHC-copper catalyst.^[35] Both a boryl group and a carboxylate group were installed to the C–C triple bonds regio- and stereoselectively, affording the α,β -unsaturated β -boralactone derivatives. The boracarboxylation with a diboron compound and CO₂ is of particular interest and importance because of the wide utility of C–B bonds in numerous chemical transformations^[36] and the challenge of catalytic regio- and stereoselective synthesis of multifunctionalized alkenes.^[37]

In this study, we started with the examination of sequential stoichiometric reactions of [(IPr)Cu(OTfBu)] with B₂(pin)₂, diphenylacetylene, and CO₂ (Scheme 10). The reaction of diphenylacetylene with [(IPr)CuB(pin)], which was instantly generated by mixing [(IPr)Cu(OTfBu)] with B₂(pin)₂,^[38] afforded the β -boryl alkenyl-copper complex **M** through *syn* insertion of the C–C triple bond into the Cu–B bond. Exposure of a benzene solution of **M** to a CO₂ atmosphere gave the carboxylation product **N** in 95% isolated yield. As disclosed by X-ray crystallographic analysis, complex **N** features a unique cyclic structure through interaction between the boron atom and the carboxylate unit. Treatment of **N** with 1.0 equivalent of LiOTfBu in THF formed quantitatively [(IPr)Cu(OTfBu)] and the corresponding lithium boralactone derivative **O** *via* transmetallation. An X-ray crystallographic study revealed that **O** possesses a dimeric structure by an ‘intermolecular’ interaction between the Li atom and the carbonyl



Scheme 10. Stoichiometric reactions of copper complexes.

group. Complexes **N** and **O** represent the first examples of structurally characterized α,β -unsaturated boralactone compounds.

In the presence of [(IPr)CuCl] (5 mol%) and LiOTfBu (1.1 equiv.), the catalytic boracarboxylation of diphenylacetylene with B₂(pin)₂ and CO₂ only yielded a trace amount of carboxylation product. When less sterically demanding [(IMes)CuCl] was used as a catalyst, a significant increase of the catalytic activity was observed, affording the boracarboxylation product in 73% isolated yield. The use of [(SIMes)CuCl], which bearing a saturated NHC ligand, further increased the

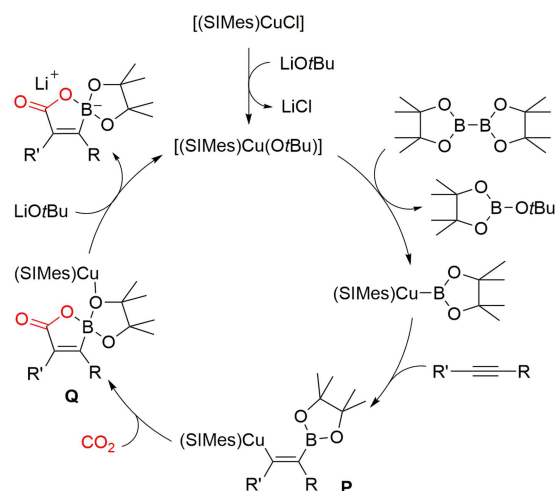
yield to 81 %. Under the optimized reaction conditions, it was found that diaryl alkynes with various substituents are generally suitable substrates, affording the expected products in good to excellent yields (Table 7). Aromatic chlorides and ester functional groups are compatible with the reaction conditions. Aromatic heterocycle substrates such as dithienylacetylene could also be converted into the corresponding product in high yield. As to unsymmetrical alkynes, the carboxylation of 1-phenyl-1-propyne and 1-phenyl-1-butyne took place exclusively at the carbon atom having the phenyl substituent to give selectively the β -phenylacrylic acid derivatives. It is noteworthy that phenylacetylene could also undergo the boracarboxylation reaction, yielding the β -phenylacrylic acid product in high selectivity. In the case of 2-*tert*-butyl-substituted phenylacetylene, only a trace amount of the carboxylation product was obtained which may be due to steric hindrance.

In a possible mechanism for the current catalytic boracarboxylation reaction (Scheme 11), the facile reaction between [(SImes)Cu(O*t*Bu)] and B₂(pin)₂ gave the copper

Table 7. Catalytic boracarboxylation of alkynes with B₂(pin)₂ and CO₂.^[a]

$\text{R}-\text{C}\equiv\text{C}-\text{R}' + \text{B}_2(\text{pin})_2 + \text{CO}_2 \xrightarrow[\text{THF, 80 }^\circ\text{C}]{\begin{smallmatrix} 5 \text{ mol\% } [(\text{SImes})\text{CuCl}] \\ \text{LiO}t\text{Bu (1.1 equiv)} \end{smallmatrix}} \text{Product}$		
	81%	
	74%	
	64%	
	77%	
	68%	
	94%	
	94%	
	83%	
	76%	
	<5%	
	71% (98:2)	

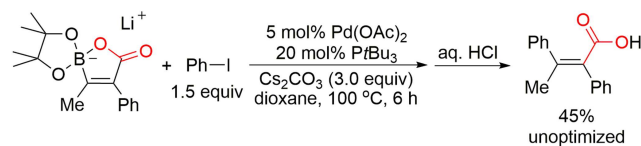
[a] Reaction conditions: [(SImes)CuCl] (5 mol%), alkyne (0.5 mmol), B₂(pin)₂ (1.0 equiv), CO₂ (1 atm), LiO*t*Bu (1.1 equiv), THF (3.0 mL), 80 °C, 14 h. Isolated yields.



Scheme 11. A possible mechanism of catalytic boracarboxylation.

boryl complex [(SImes)CuB(pin)]. The following *syn* insertion of an alkyne into the Cu–B bond generated the β -borylalkenyl copper complex **P**. The subsequent addition of **P** to CO₂, followed by replacement of the NHC-copper moiety in the resulting carboxylate with the boron atom yielded the corresponding β -boralactone derivative **Q**. Finally, [(SImes)Cu(O*t*Bu)] was regenerated *via* the transmetalation between the copper complex **Q** and LiO*t*Bu while releasing the final boracarboxylated product.

The synthetic utility of the lithium α,β -unsaturated boralactone products was proved by a Suzuki-Miyaura cross-coupling reaction with iodobenzene, which afforded the tetra-substituted alkene in 45 % isolated yield (Scheme 12).

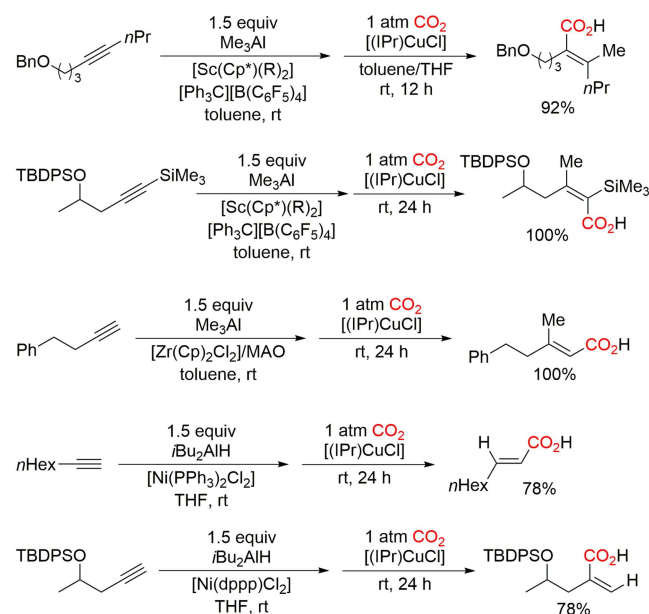


Scheme 12. An example of Suzuki-Miyaura cross-coupling reaction.

5.2. Formal Methylation and Hydrogenative Carboxylation of Alkynes

As mentioned previously, the alkylative^[39] or hydrogenative^[40] carboxylation of alkynes with CO₂ represent as an efficient method for the synthesis of α,β -unsaturated carboxylic acids. On the other hand, regio- and stereospecific hydroalumination and methylalumination of C–C triple bonds have been reported.^[41] In these reactions, the alkenylaluminum species

can be prepared with well-controlled configurations by choosing an appropriate catalyst. By combination of such catalytic methylalumination or hydroalumination of alkynes with the Cu-catalyzed carboxylation of the resulting alkenylaluminum species with CO₂, we have developed the regio- and stereospecific formal methylative and hydrogenative carboxylation of alkynes with CO₂ (Scheme 13), affording a variety of



Scheme 13. One-pot carboxylation of various alkynes with CO₂.

α,β -unsaturated carboxylic acids which were difficult to prepare previously.^[42]

In the presence of 5 mol% of [(IPr)CuCl], the carboxylation of the alkenylaluminum, which was *in situ* prepared by [Sc(CH₂C₆H₄NMe₂-*o*)₂(Cp*)]/[Ph₃C][B(C₆F₅)₄]-catalyzed methylalumination of an internal alkyne containing a tethered ether group,^[41g] proceeded smoothly to give the corresponding carboxylic acid product in high yield. The directing group led the carboxylation to take place at the proximal position and the methylation at the distal position in a *cis* manner.

It has been reported that Sc-catalyzed methylalumination of trimethylsilyl-substituted alkynes containing a tethered ether group gives the alkenylaluminum species in *anti*-configuration.^[41g] By combination of this *anti*-methylalumination with Cu-catalyzed carboxylation, the sequential *anti*-selective methylalumination/carboxylation was also achieved, providing (*E*)- α -silyl- β -methyl- α,β -unsaturated carboxylic acid through introducing the carboxyl group to the distal position and the methyl group to the proximal position from the ether group.

In the case of terminal alkynes, the methylalumination could be achieved regiospecifically in the presence of Zr catalyst.^[41f] The [(IPr)CuCl]-catalyzed carboxylation of the resulting alkenylaluminum with CO₂ occurred efficiently to introduce the carboxyl group to the less hindered C1-position and the methyl group to the C2-carbon atom in a *syn* fashion.

By combining [NiCl₂(PPh₃)₂]-catalyzed hydroalumination of terminal alkynes^[41b] and Cu-catalyzed carboxylation, the formal hydrocarboxylation of 1-octyne proceeded to afford (*E*)- α,β -unsaturated carboxylic acid while the alumination/carboxylation took place at the C1-position and the hydride was added to the C2-position in a *cis* manner. When [NiCl₂(dppp)] was used,^[41b] the aluminum/carboxylation occurred at the C2 position rather than C1 position, giving α,β -unsaturated- β -carboxylic acids.

5.3. Alkylative Carboxylation of Ynamides

The carboxylation of ynamides, which have been extensively studied as functionalized alkyne derivatives in various chemical transformations,^[43] can afford structurally interesting α,β -dehydro- α -amino acids.^[44] We have developed a NHC-copper catalyzed regioselective alkylative carboxylation of ynamides using CO₂ and dialkylzinc reagents, which provide α,β -dehydro- α -amino acid derivatives bearing two different β -substituents with well-controlled configuration.^[45]

In the presence of a catalytic amount of [(IPr)CuCl], the three-component coupling of ynamides, dialkylzinc reagents, and CO₂ provided the alkylative carboxylation products in high yields after hydrolysis and esterification (Table 8). Besides Me₂Zn, both Et₂Zn and Bu₂Zn are also suitable for this reaction without the observation of β -hydride elimination, which is often observed in the related catalyst systems.^[44] In all cases, the reactions took place in a *syn* manner, introducing the alkyl group to the β -position of the ynamides and the carboxyl group to the α -position selectively. Ynamides bearing 2-oxazolidinone, pyrrolidine-2-one, and Evans' chiral oxazolidin-2-one moieties were found to be good substrates in this catalytic alkylative carboxylation reaction, affording the desired products in high yields when different kinds of dialkylzinc reagents were used. It was also found that acyclic Boc-protected ynamides are suitable substrates for ethylative and butylative carboxylation with Et₂Zn and Bu₂Zn, giving the desired products in good yields, while Me₂Zn is not applicable to these acyclic substrates.

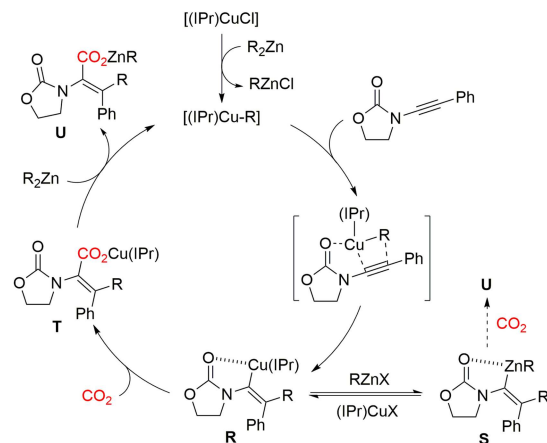
A possible reaction mechanism for this copper-catalyzed alkylative carboxylation of ynamides with CO₂ and dialkylzinc reagents is illustrated in Scheme 14. The initial transmetalation between [(IPr)CuCl] and R₂Zn would form an alkylcopper species [(IPr)CuR]. The carbonyl group of ynamide works as a directing group in the following carbocupration step, generating the alkenylcopper intermediate **R** regioselectively.

Table 8. One-pot alkylative carboxylation of ynamides.^[a]

$\text{EWG}-\text{N}(\text{R}^2)\text{C}\equiv\text{C}-\text{R}^1 \xrightarrow[\text{THF, 50 } ^\circ\text{C}]{\begin{smallmatrix} 1\text{atm CO}_2 \\ 1.5\text{ equiv R}_2\text{Zn} \\ 5\text{ mol\% }[(\text{IPr})\text{CuCl}] \end{smallmatrix}} \xrightarrow[\text{2. CH}_2\text{N}_2]{\text{1. H}_3\text{O}^+} \text{EWG}-\text{N}(\text{R}^2)\text{C}(\text{CO}_2\text{Me})=\text{C}(\text{R})\text{R}^1$		
Ynamides	R ₂ Zn	Product
	Me ₂ Zn	93 %
	Et ₂ Zn	92 %
	Bu ₂ Zn	92 %
	Me ₂ Zn	96 %
	Et ₂ Zn	96 %
	Bu ₂ Zn	88 %
	Me ₂ Zn	96 %
	Et ₂ Zn	97 %
	Bu ₂ Zn	94 %
	Me ₂ Zn	83 %
	Et ₂ Zn	83 %
	Bu ₂ Zn	85 %
	Me ₂ Zn	97 %
	Et ₂ Zn	90 %
	Bu ₂ Zn	80 %
	Et ₂ Zn	76 %
	Bu ₂ Zn	75 %
	Et ₂ Zn	79 %
	Bu ₂ Zn	83 %

[a] Isolated yield.

Transmetalation between **R** and another R₂Zn may form an alkenylzinc species **S**, which could regenerate **R** via a reversible process. Although a pathway through direct nucleophilic



Scheme 14. A possible mechanism of catalytic alkylative carboxylation.

addition of **S** to CO₂ cannot be ruled out, we currently speculate that the carboxylation would be achieved by the insertion of CO₂ into the Cu–C bond in **R**, affording the corresponding copper carboxylate **T**. Finally, transmetalation between **T** and R₂Zn would yield zinc carboxylate **U** and regenerate active species [(IPr)CuR].

Compared to dialkylzinc reagents, the use of alkylzinc halides (RZnX) is more desirable because various alkylzinc halides can be easily prepared from the corresponding alkyl halides and metallic zinc.^[46] Moreover, the higher functional group tolerability of alkylzinc halides^[47] might enable the introduction of functionalized alkyl groups during the carboxylation process. We found that Cu(acac)₂, instead of [(IPr)CuCl], can efficiently catalyze the carboxylation of ynamides with RZnX. By the combination of this carboxylation with the carboxylation of the resulting alkenyl zinc species, the formal alkylative carboxylation of ynamides with RZnX and CO₂ has been achieved (Table 9).^[48] The addition of LiCl is indispensable for the carboxylation process.^[49] The silyl ether, ester, and terminal alkenyl groups on alkylzinc halides were tolerated to give the corresponding alkylative carboxylation products in good yields.

5.4. Alkylative Carboxylation of Allenamides

Previously, a nickel-mediated carboxylation of allenamides with CO₂ have been reported to give α,β-dehydro-β-amino acid derivatives, which requires a stoichiometric amount of Ni(cod)₂.^[50] The development of catalytic version of the carboxylation of allenamides with CO₂ is of interest.^[51] We have achieved a [(IPr)CuCl]-catalyzed alkylative carboxylation of allenamides with CO₂ and dialkylzinc reagents, affording α,β-dehydro-β-amino acid esters with high regio- and stereoselectivity.^[52]

Table 9. Formal alkylative carboxylation of an ynamide with alkylzinc halides.

Entry	RZnX	Product	Yield [%] ^[a]
1	BuZnI		78
2	TBDMSO(CH2)4ZnI		76
3	PhCOO(CH2)4ZnI		79
4	EtOCO(CH2)3ZnI		77 ^[b]
5	CH2=CH(CH2)3ZnBr		72

[a] Isolated yields. [b] The reaction was carried out by using 35 mol% of Cu(acac)₂ and the carboxylation process was performed at 50 °C.

The methylative carboxylations of allenamide substrates were carried out in a one-pot manner by introducing CO₂ (1 atm) into a THF solution of allenamide, Me₂Zn (1.5 equivalent), and [(IPr)CuCl] (5 mol%). The desired methylative carboxylation products were obtained in high yields after hydrolysis and esterification (Table 10). Besides Me₂Zn, both Et₂Zn and Bu₂Zn were found to be suitable alkylating reagents in this reaction without the observation of β-hydride elimination.

It was found that allenamides bearing 2-oxazolidinone moieties reacted efficiently with different dialkylzinc reagents, giving the corresponding alkylative carboxylation products in high yields. Allenamides containing different ring size lactam units are also suitable substrates for the alkylative carboxylation reaction. Even acyclic allenamides are applicable for this method. The desired products were obtained in moderate yields when the reactions were carried out at room temperature.

Similarly, the formal alkylative carboxylation of allenamides with alkylzinc halides and CO₂ has been achieved by combining the Cu(acac)₂-catalyzed carboxylation and the LiCl-promoted carboxylation (Table 11).^[48] Alkylzinc halides bearing functional group, such as silyl ether, ester, nitrile, and terminal alkenyl groups, are applicable to afford the corresponding alkylative carboxylation products in good yields.

6. Boracarboxylation of Aldehydes

Due to its thermodynamic stability and kinetic inertness, the chemical transformations of CO₂ for practical synthetic applications remain very limited. Among the most successful transformations of CO₂ for chemical synthesis, the cyclic addition of CO₂ to epoxides for the synthesis of cyclic carbonates bearing various exocyclic functional substituents have been extensively studied.^[53] On the other side, installation of a heteroatom into a cyclic carbonate framework could be a promising way for the construction of a novel type of heteroatom-containing cyclic carbonate skeleton. However, such kind of reactions has not been achieved previously, which may due to the difficulty in suppressing possible competing side reactions in a multi-component reaction system.

Electrolyte is a key component for the construction of lithium-based rechargeable batteries.^[54] Due to their unique properties, functional lithium borate salts such as lithium bis(oxalato)borate (LiBOB) have been intensively studied as electrolytes in high performance lithium batteries.^[55] The development of efficient and versatile chemical transformations for the preparation of diverse functional lithium borate salts from easily available starting materials is highly desirable due to their limited availability. On the other hand, in lithium ion batteries, cyclic carbonates such as propylene carbonate and ethylene carbonate are frequently used as electrolyte solvents.^[54] The electrolyte solution composed of lithium borate electrolyte and carbonate solvent enables the formation of solid electrolyte interphase (SEI) on electrode surfaces, which has been proved to be critical for improving the performance.^[54] Considering the importance of the above mentioned facts, we are interested in the preparation of lithium borate salts containing both a cyclic carbonate moiety and a borate unit in one molecule by the use of CO₂ as a starting material.

We have accomplished the first one-pot multi-component coupling reaction of CO₂, B₂(pin)₂, LiOtBu, and various aldehydes, enabling highly efficient and selective preparation of novel lithium cyclic boracarbonate ion pair compounds which feature a unique cyclic carbonate structure fused with a pinacolatoborate moiety.^[56] We examined various NHC-copper catalysts for the reaction of benzaldehyde with B₂(pin)₂ (1.0 equivalent) and LiOtBu (1.1 equivalent) under a CO₂

Table 10. One-pot alkylative carboxylation of allenamides.^[a]

$\text{EWG}-\text{N}(\text{R}^1)-\text{C}\equiv\text{C}-\text{CH}_2-\text{CHO} \xrightarrow[\text{THF, 50 } ^\circ\text{C}]{\begin{smallmatrix} 1 \text{ atm CO}_2 \\ 1.5 \text{ equiv R}_2\text{Zn} \\ 5 \text{ mol\% [(IPr)CuCl]} \end{smallmatrix}} \xrightarrow[\text{2. CH}_2\text{N}_2]{1. \text{ H}_3\text{O}^+} \text{EWG}-\text{N}(\text{R}^1)-\text{C}(\text{R})=\text{CH}-\text{CO}_2\text{Me}$		
Ynamides	R ₂ Zn	Product
	Me ₂ Zn Et ₂ Zn Bu ₂ Zn	 82 % 79 % 78 %
	Me ₂ Zn Et ₂ Zn Bu ₂ Zn	 80 % 82 % 72 %
	Me ₂ Zn Et ₂ Zn Bu ₂ Zn	 82 % 78 % 76 %
	Me ₂ Zn Et ₂ Zn Bu ₂ Zn	 61 % 78 % 76 %
	Me ₂ Zn Et ₂ Zn Bu ₂ Zn	 41 % ^[b] 64 % ^[b] 55 % ^[b]

[a] Isolated yield. [b] Reaction was carried out at rt.

atmosphere. A lithium cyclic boracarbonate ion pair product **V** was detected when the reaction was carried out under 1 atm of CO₂ with [(IPr)CuCl] as a catalyst, albeit only in a trace amount. In the presence of [(SiMe)₃CuCl] catalyst, raising the CO₂ pressure to 5 atm led to the isolation of **V** in 85 % yield (Table 12). X-ray crystallographic study of single crystals from recrystallization of **V** revealed that it adopts a dimeric structure of two novel boracarbonate units. The unique boracarbonate ring is built up by connection of the two oxygen atoms of the carbonate group across a B–C bond.

The scope of aldehydes for the present coupling reaction with CO₂ was then investigated under the optimized reaction conditions (Table 12). A wide range of aromatic aldehydes bearing either electron-donating or electron-withdrawing

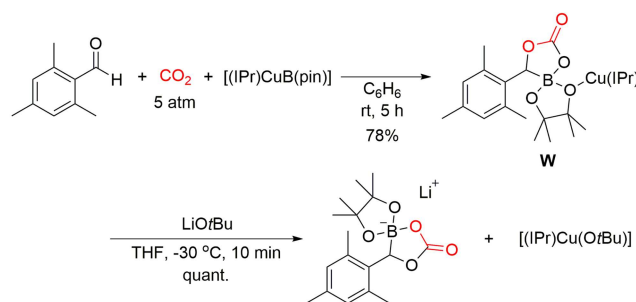
Table 11. Alkylative carboxylation of an allenamide with various alkylzinc bromides.

$\text{Oxazolidinone}-\text{C}\equiv\text{C}-\text{CH}_2-\text{CHO} \xrightarrow[\text{DMF, 50 } ^\circ\text{C}]{\begin{smallmatrix} 2 \text{ equiv RZnBr} \\ 2.5 \text{ equiv LiCl} \\ 35 \text{ mol\% Cu(acac)}_2 \end{smallmatrix}} \xrightarrow[\text{then CH}_2\text{N}_2]{\begin{smallmatrix} 1 \text{ atm CO}_2 \\ 50 ^\circ\text{C (one-pot)} \end{smallmatrix}} \xrightarrow{\text{H}_3\text{O}^+} \text{Oxazolidinone}-\text{C}(\text{R})=\text{CH}-\text{CO}_2\text{Me}$			
Entry	RZnBr	Product	Yield [%] ^[a]
1	BuZnBr		83
2			68
3	TIPSO-(CH ₂) ₃ -ZnBr		71
4	EtO-C(=O)-(CH ₂) ₅ -ZnBr		76
5	NC-(CH ₂) ₃ -ZnBr		80

[a] Isolated yields.

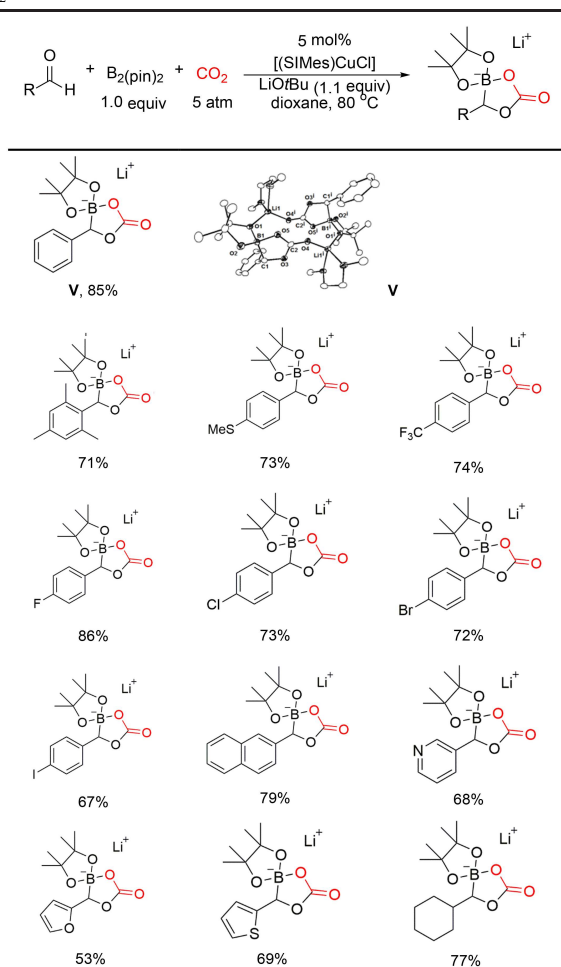
groups are able to afford the corresponding products efficiently. Even the reaction of sterically demanding mesitaldehyde occurred smoothly to give the desired product in 71 % isolated yield. The halides substituted benzaldehydes are suitable for this reaction, affording the corresponding halogenated products in good yields. Heteroaromatic aldehydes are also compatible with the reaction conditions, efficiently affording the desired products. Besides aromatic aldehydes, various aliphatic aldehydes are also suitable substrates for the current reaction, giving the corresponding products in high yields.

The stoichiometric reaction of [(IPr)CuB(pin)] with mesitaldehyde under 5 atm of CO₂ was examined, from which a cyclic boracarbonate complex **W** containing a (IPr)Cu unit was isolated in 78 % yield (Scheme 15). Treatment of **W** with



Scheme 15. Stoichiometric reactions of copper complexes.

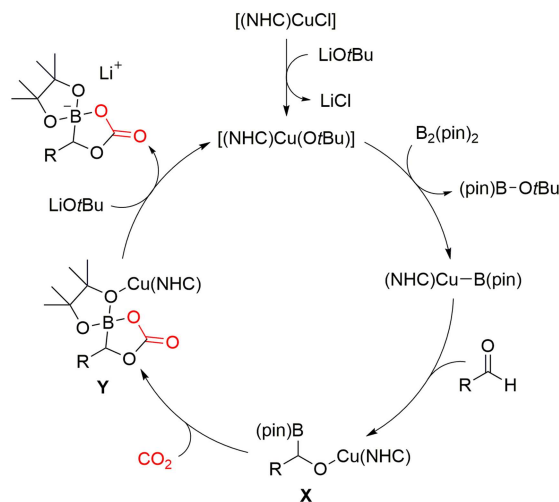
Table 12. Cu-catalyzed coupling of aldehydes with B₂(pin)₂ and CO₂.^[a]



[a] Reaction conditions: [(SIMes)CuCl] (5 mol%), B₂(pin)₂ (0.5 mmol), aldehyde (0.5 mmol), LiOtBu (1.1 equiv), dioxane (3.0 mL), CO₂ (5 atm), 80 °C, 20 h. Isolated yields.

1 equivalent of LiOtBu in THF afforded the corresponding lithium ion pair product and [(IPr)Cu(OtBu)] quantitatively.

In a possible mechanism for the current catalytic multi-component coupling reaction (Scheme 16), the insertion of an aldehyde into the Cu–B bond of [(NHC)CuB(pin)] gave the copper alkoxide **X**. The subsequent insertion of CO₂ into the newly formed Cu–O bond, followed by migration of the NHC-copper unit from the resulting carbonate group to a pinacolate oxygen atom and intramolecular B–O bond formation generated the cyclic boracarbonate intermediate **Y**. Transmetalation between copper complex **Y** and LiOtBu regenerated active species [(NHC)Cu(OtBu)] and released the desired lithium boracarbonate ion pair product.



Scheme 16. Proposed reaction mechanism for the coupling of aldehydes with B₂(pin)₂ and CO₂.

Remarkably, this multi-component coupling reaction took place in a highly selective and efficient fashion. Several side reactions were successfully suppressed. From easily available starting materials, a new class of lithium cyclic boracarbonate ion pair compounds were synthesized, which are of interest as potential electrolyte candidates for lithium ion batteries in view of their unique structural features.

7. Summary and Outlook

Various types of transformations using CO₂ as an attractive and sustainable C1 source have been achieved in high efficiency and selectivity by employing NHC-copper complexes as versatile catalysts. The unique steric and electronic properties of these catalysts enabled the facile insertion of CO₂ into organocopper intermediates which were generated *via* different types of selective processes such as transmetalation, C–H bond activation, carbocupration, and borylcupration. Efficient transmetalation of organoboron compounds or *in situ* formed organoaluminum species to copper center of the catalyst is crucial to the carboxylation of organoboron reagents, formal C–H carboxylation of aromatics and allyl aryl ethers, formal methylative and hydrogenative carboxylation of alkynes. The direct carboxylation of heterocyclic aromatics involves the activation of C–H bonds for the formation of arylcopper intermediate. The alkylative carboxylation of ynamides and allenamides has been achieved by selective carbocupration of C–C multiple bonds with alkylzinc reagents. The new lithium cyclic borolactone or boracarbonate compounds were synthesized from boracarboxylation reactions of alkynes or aldehydes, which were accomplished *via* highly

selective borylcupration/ CO_2 insertion/cyclization sequence. The accelerated insertion of CO_2 into the NHC-copper hydride is the key step in efficient hydrosilylation of CO_2 . In these catalytic processes, NHC ligands stabilize the active copper species which enabled the isolation of different key intermediates, providing important insights for understanding the reaction mechanisms. These studies may inspire the development of new catalytic systems and the design of new catalytic transformations for the CO_2 utilization.

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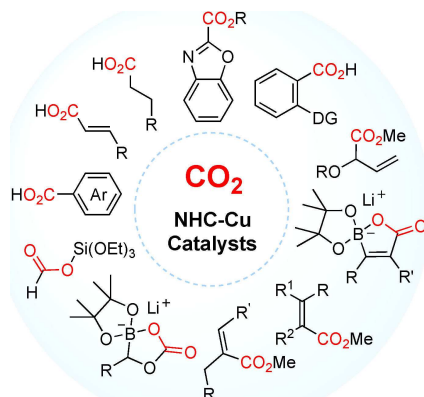
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PERSONAL ACCOUNT

A series of carboxylation reactions with CO_2 , such as carboxylation of organo-boron reagents and C–H bonds, hydrogenative and alkylative carboxylation of alkynes and allenes, boracarboxylation of alkynes and aldehydes, hydrosilylation of CO_2 , have been developed by using N-heterocyclic carbene-copper catalysts. Isolation of key intermediates provides important insights for the reaction mechanisms.



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Carboxylation Reactions with Carbon Dioxide Using N-Heterocyclic Carbene-Copper Catalysts